Rojina Adhikari

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Education

Massachusetts Institute of Technology Cambridge, MA
B.S. in Computer Science/Artificial Intelligence (6-4) & Mathematical Economics (14-2) May 2029
Coursework: Machine Learning, CS & Programming using Python, Math for CS, Microeconomics, Macroeconomics, Multivariate Calculus, Linear Algebra, LLMs from Scratch, Statistical Methods in Economics, Organizational Economics
Activities: IEEE/ACM, Society of Women Engineers, Undergraduate Society of Women in Math, Bitcoin Club Logistics, Career Exploration Leader, Alpha Chi Omega (Chapter Relations & Standards, Philanthropy, Recruitment Committee)

Work Experience

MIT Economics Department Cambridge, MA
AI Engineering Researcher August 2026 – Present
- Develop LLM/NLP pipelines for a 17M+ document policy corpus, using Qwen 3 8B to extract structured policy records with source-level provenance.
- Built LLM validation and Jina embedding-based classification pipelines, achieving 99.4% LLM-judged semantic support across 1,474 extracted policies from 25 documents.

MIT linQ at Institute for Medical Engineering and Sciences Cambridge, MA
Data Science & Outcomes Researcher September 2026 – Present
- Engineer data pipelines and relational data models using SQL, Snowflake, and Airtable to integrate, clean, and standardize multi-source healthcare innovation data for longitudinal outcomes analysis and real-time reporting.

MassHealth Analytics Boston, MA
Data & AI Engineering Intern June 2026 – August 2026
- Built and deployed a Python/Selenium ETL pipeline integrating GitHub Copilot to extract and standardize Tableau metadata and generate AI-assisted dashboard documentation for analysts across 12 Massachusetts agencies.
- Implemented components of AWS pipeline using Bedrock, Lambda, and Aurora PostgreSQL to automate document extraction, reconcile conflicting records, and enable natural-language search.

MIT Senseable City Lab Cambridge, MA
Data Science & AI Undergraduate Researcher September 2025 – June 2026
- Developed Python simulation pipelines to generate and evaluate 1M+ constraint-compliant housing configurations for large-scale urban modeling experiments.
- Investigated GeoAI model generalizability across spatial partitions, analyzing how geographic proximity, feature similarity, and underlying data distributions affect predictive transferability.

i2E, Inc. Oklahoma City, OK
Data Science Intern April 2024 – May 2025
- Designed computational models to analyze operational and financial data for 10+ startups, supporting investment diligence and evaluating automation opportunities.
- Modeled time-series workflow and labor-allocation data across business functions, identifying process bottlenecks and improving workplace efficiency by 20%.

University of Tulsa Center for Real Estate Studies Tulsa, OK
Economics Researcher April 2024 – October 2024
- Applied GIS mapping and regression analysis to examine spatial patterns in real estate investment and urban growth, translating findings into 3 actionable policy recommendations.

Leadership & Projects

Software Developer, Adaptive Gaming Controller Cambridge, MA
MIT Assistive Technology Club September 2025 – Present
- Developed multimodal AI assistant integrating OpenAI vision/speech models, OpenCV, and Piper TTS to interpret live gameplay and translate voice commands into controller inputs.
- Integrated screen capture, AI inference, and speech processing through Python/FastAPI to create an accessible gaming system for users with limited mobility.

Flow State | MIT web.lab Programming Competition January 2026
- Built full-stack MERN study-space discovery platform with RESTful APIs, MongoDB data models, Google OAuth authentication, and persistent user reviews/bookmarks.
- Implemented interactive geospatial discovery using React Leaflet, featuring map-based navigation and dynamic location filtering.

Rola, Co-Founder and Software Engineer | MIT Sandbox Innovation Fund January 2026 – Present
- Engineer full-stack peer-to-peer services marketplace using MERN, implementing user authentication, service listings, search, and transaction workflows.

Skills

Languages & Libraries: Python, SQL, R, Java, JavaScript, TypeScript, HTML/CSS, MATLAB, Pandas, NumPy, PyTorch
Frameworks & Tools: React, Node.js, Express.js, Selenium, Snowflake, Git, AWS, MongoDB, Tableau, Figma, Linux/Bash